

ColorFlow:

An Approach to Colorized Pixel Representation in EmerFlux

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Abstract—In this study, we propose a method for generating multi-colored protrusions on a black oil-covered water surface. The method combines multiple liquids to produce these protrusions. In the proposed system, liquids of different colors are supplied by two pumps, combined into a single nozzle through a T-shaped junction, and then injected onto the water surface. By controlling the flow rate ratio of the pumps, the color of a single pixel can be continuously adjusted.

Experimental results showed that color representation depends on the viscosity of the liquid. Under high-viscosity conditions, colors remained separated, whereas under low-viscosity conditions, colors mixed and changed continuously. These results indicate that the proposed method enables the addition of color information to water-surface displays, allowing richer and more expressive information presentation.

Keywords—Liquid-based display, Shape-changing interface, Color mixing, Water surface

I. INTRODUCTION

Liquids possess properties such as fluidity and continuous shape transformation. These characteristics make them a promising medium for visual expression and interaction. In recent years, numerous studies and media art works have explored the use of liquids for information display and expression by controlling their state and shape. These approaches enable new visual experiences characterized by continuous and organic changes.

We have previously developed a water-surface display system called EmerFlux, which utilizes a two-layer structure of water and oil. In this system, white-colored water in the lower layer is pushed upward. As a result, circular white pixels are formed on the surface of black oil in the upper layer. This structure enables clear visual contrast between the background and the pixels.

However, each pixel in EmerFlux is generated using a single liquid. Therefore, multiple colors cannot be represented within a single pixel. Color plays an important role in information recognition and meaning. Thus, this limitation restricts the expressive capability of the display.

In this study, we propose a method to control pixel color by mixing liquids of different colors. In the proposed system, two pumps independently supply different colored liquids. These liquids are merged into a single nozzle using a T-shaped junction. By adjusting the flow rate ratio of the pumps, the liquids are mixed. The mixed liquid is then injected onto the

water surface. As a result, a single pixel can represent a variety of colors.

II. RELATED WORK

Research on liquid-based information display includes interactive systems that use the water surface as a display medium. AquaTop Display [1] projects images onto a water surface and enables gesture-based interaction. This approach leverages the fluidity and tactile properties of water. However, because it relies on projection, it is sensitive to ambient lighting conditions and visibility.

Another approach uses changes in the state of liquids for display. Condensation-based displays generate images by controlling the formation of water droplets [2]. In this method, grayscale representation is achieved by adjusting the density and distribution of droplets. However, this approach is limited to grayscale and cannot represent multiple colors.

In this study, we propose a method for controlling pixel color by mixing liquids of different colors. The proposed method utilizes the physical deformation of liquids and enables continuous color control. By integrating this approach with EmerFlux, it extends the expressive capabilities of water-surface displays.

III. SYSTEM IMPLEMENTATION

The system uses three pumps: two for color mixing and one for drainage. These pumps are controlled by an Arduino Mega via a transistor array (TBD62783APG). A 12 V power supply is connected to the transistor array. The driving voltage of each pump is adjusted using PWM signals from the Arduino. This configuration enables independent control of the flow rate of each pump.

The two-colored liquids are supplied from separate reservoirs. These reservoirs are made of 3 mm thick acrylic plates fabricated by laser cutting. The display tank, where pixels are generated, is also constructed from acrylic plates and contains a layer of oil. The mixed liquid is injected into this tank to form pixels on the liquid surface.

The water level in the display tank is maintained using a drainage pump, ensuring stable display conditions. Silicone tubes with an inner diameter of 6 mm and an outer diameter of 8 mm are used to transport liquids between the tanks.

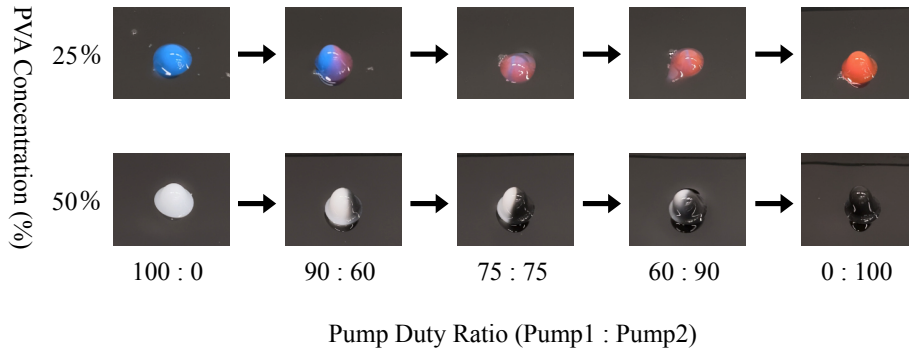


Fig. 3. Color Transition under Different PVA Concentrations and Pump Duty Ratios

To stabilize the shape of the injected liquid, polyvinyl alcohol (PVA) is mixed with water. In this study, the PVA solution is colored with acrylic paint to generate liquids of different colors. Figure 1 shows the system configuration of the proposed color mixing mechanism. Figure 2 shows the appearance of the implemented device.

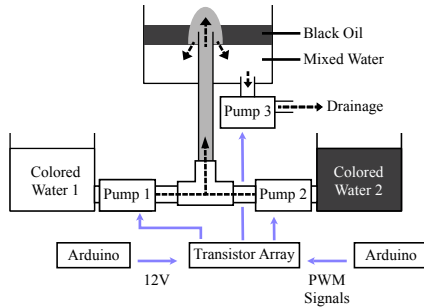


Fig. 1. System Configuration of the Liquid Mixing and Pump Control System

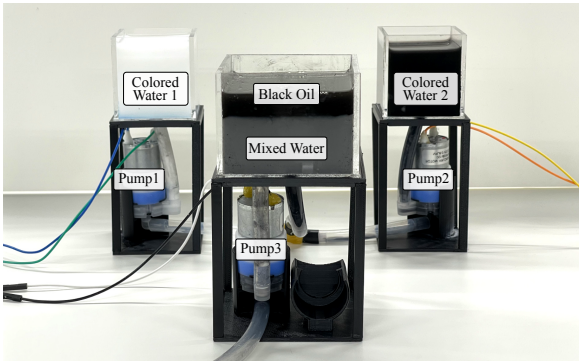


Fig. 2. Appearance of the implemented device.

IV. RESULTS

In this experiment, we investigated the effect of liquid viscosity on color representation by varying the mixing ratio of PVA and water. Two conditions were tested: approximately 50% and 25% PVA concentrations. Two color pairs were used: black and white, and red and blue. The black–white pair was selected based on the original EmerFlux configuration. The red–blue pair was used to evaluate color mixing behavior.

The pumps were controlled using PWM signals with values above 60%, as stable fluid injection was not observed below this threshold. During the experiment, the mixing ratio of the liquids was controlled by dynamically adjusting the output ratio of the two pumps over time.

Under the 50% PVA condition, the two colors remained clearly separated within a single pixel. The area occupied by each color changed according to the pump output ratio. This behavior is attributed to the high viscosity of the liquid, which prevented sufficient mixing inside the tube.

In contrast, under the 25% PVA condition, the two colors were mixed before being injected onto the water surface.

In this case, the color changed continuously according to the pump ratio. These results are shown in Fig. 3.

V. CONCLUSION

In this study, we proposed a method for extending color representation in liquid-based displays by controlling pixel color through the mixing of multiple liquids. A two-pump mixing mechanism was implemented, enabling color variation within a single pixel by adjusting the flow rate ratio.

Experimental results showed that color representation varies depending on the viscosity of the liquid. Under high-viscosity conditions, colors remained separated, whereas under low-viscosity conditions, colors mixed and changed continuously. These results demonstrate that the proposed method enables switching between different modes of color expression.

As future work, the system can be extended to a grid structure to enhance information presentation on the water surface. By assigning multiple colors to each pixel and controlling their distribution, smoother and more continuous representations can be achieved compared to binary on/off pixel displays. In addition, the system enables the display of different information depending on the viewing angle. This allows for multi-perspective representations, similar to interactive artworks such as “Penguin Mirror” by Daniel Rozin [3].

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